Problem A. A Problems and A Balloons

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

In the ICPC competitions, the quota of a region in the Superregional depends, among other things, on the number of teams in the regional competition that have solved at least one problem.

When a team solves a problem, it receives a balloon. At this point, the team cannot submit solutions for a problem that has already been solved (meaning each problem can only be solved once by a team).

There are n teams participating in the competition. A total of k balloons have been distributed. What is the maximum number of teams that could have solved at least one problem?

Input

The first line of input contains a single integer n ($1 \le n \le 1000$) — the number of teams participating in the competition. The second line of input contains a single integer k ($1 \le k \le 1000$) — the number of distributed balloons.

Output

Output a single integer — the maximum possible number of teams that could have solved at least one problem given the number of teams and distributed balloons.

Examples

standard input	standard output
20 25	20
11 2	2

Problem B. Big Alphanumeric String

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

Given a string consisting of decimal digits.

You plan to rearrange the digits so that the number of pairs of adjacent differing digits is minimized. For example, in the string 02112025, the number of pairs of adjacent differing digits is 6 (02, 21, 12, 20, 02, 25), and in 0101 there are 3 (01, 10, 01).

How many ways are there to rearrange the digits in the required manner? The ways are considered different if the resulting strings from the rearrangement are different.

Input

The input consists of a string made up of digits from 0 to 9 and having a length of no more than 1000 characters.

Output

Output a single integer — the number of ways.

Example

standard input	standard output
2025	6

Problem C. Count The Sum Of GCD

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5 seconds
Memory limit:	1024 megabytes

You have two numbers $L \leq R$, defining the set of all natural numbers from L to R, and a natural number X. Count the number of triples c, g, s such that:

- There exists a non-empty subset of the given set, the sum of whose elements equals s, the number of elements in which is c, and the greatest common divisor of the elements is g.
- The equality $c \cdot g + s = X$ holds.

Input

In the first line, you are given a number t ($1 \leq t \leq 10$), denoting the number of test cases. In the next t lines, you are given three numbers L, R, X, where $1 \leq L \leq R \leq 10^{10}$ and $1 \leq X \leq 10^{18}$, defining the range of elements in the set and the parameter of the equation.

Output

For each test case, output a single number on a separate line – the number of triples that satisfy the conditions.

Examples

standard input	standard output
5 3 14 17 1 12 12 88 114 305 78 103 206 1 26 100	2 4 2 3 16
6 6 10 30 3 14 17 1 12 12 88 114 305 78 103 206 1 26 100	2 2 4 2 3 16

Problem D. Death Star N

Input file:	standard input
Output file:	standard output
Time limit:	1 second
Memory limit:	1024 megabytes

In a galaxy far, far away, rebels are preparing to attack another Death Star. The Death Star is located at the origin (i.e., at the point with coordinates $(0, 0, 0)$). The defense system of the Death Star works as follows: if three ships are in the same plane as the Death Star, the ships are immediately destroyed.

Darth Vader managed to intercept a message from the rebels containing the coordinates of three distinct points where the ships are supposed to arrive. The coordinates are integers. Using special transmitters, Darth Vader can introduce distortions: adding an arbitrary fractional part to each number. For example, he can turn the number 0 into 0.5 or $1/\sqrt{2}$ (but cannot make it, for example, -0.3), from the number -3 he can make it $-\pi$, from the number 2 he can make it e, and so on. Introducing distortions to one number requires energy expenditure, so Darth Vader is interested in the minimum number of coordinates among the nine transmitted coordinates that he needs to distort in order for the rebel ships to be exposed to the Death Star, and in how many ways he can choose the numbers for distortion. Two ways are considered different if there exists a coordinate that was distorted in one of these ways and was transmitted unchanged in the other.

Input

The input consists of three lines, each containing three integers. The first line contains the coordinates x_1, y_1, and z_1 of the first ship, the second line contains the coordinates x_2, y_2, and z_2 of the second ship, and the third line contains the coordinates x_3, y_3, and z_3 of the third ship ($-1000 \le x_i, y_i, z_i \le 1000$).

Output

Output two integers—the minimum number of coordinates that Darth Vader must distort to achieve his goal, and the number of ways to choose the corresponding set of coordinates. If Darth Vader cannot achieve his goal, output a single integer -1.

Examples

standard input	standard output
1 1 2 2 3 4 0 0 5	1 1
1 1 2 2 3 4 0 5 0	0 1
1000 0 0 0 -1000 0 0 0 1000	-1

Note

In the first example, adding 0.5 to the second coordinate allows the three points—the first, the second, and the "Death Star— to lie on the same line (and four of them to lie in the same plane). In the second example, the ships are already exposed, so no changes are needed (and obviously, there is only one way to "change nothing"). In the third example, the plane in which the ships are located is too far from the Death Star for Darth Vader to succeed in his plan.

Problem E. Exclusions

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

You have an array A of n non-negative integers. Consider its partition into consecutive blocks, where the cost of a block is defined as the *mex* of the elements in that block, and the cost of the partition is the minimum of the costs of the blocks.

Note that *mex* (minimal excluded) of the set is the minimal non-negative integer that does not present in the set.

Let $f(k)$ denote the maximum possible cost of partitioning the array A into k blocks. Your task is to find $f(1), f(2), f(3), ..., f(n)$.

Input

The first line contains the number of test cases T ($1 \le T \le 2 \cdot 10^5$). Following this, the test cases are provided. Each test case starts with a number n ($1 \le n \le 2 \cdot 10^5$). The next line contains the array A of n non-negative integers a_i ($0 \le a_i \le 10^9$). It is guaranteed that the sum of n across all test cases does not exceed $4 \cdot 10^5$.

Output

For each test case, output n numbers on a separate line – $f(1), f(2), ..., f(n)$.

Example

standard input	standard output
4 1 0 5 0 1 2 1 0 3 1 1 0 8 3 0 1 0 0 4 2 1	1 3 2 0 0 0 2 0 0 5 2 1 0 0 0 0 0

Problem F. Fix The Leak!

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

Princess Leia received the information that Darth Vader got a leak about some secret group of rebels that infiltrated into the technical department of the Galactical Empire.

The group consists of n agents. Each agent is assigned a number from 0 to $n - 1$. The head of the group has number 0 and is controlled directly by jedi. For each of the other agents, there is a unique group member with a number less than that of this member to whom they pass information (this can be either the head of the group or another group member). Group members who do not receive information from anyone are called spies.

When a command to stop the work is received by a certain agent, they cease intelligence activities, and the entire chain of agents supplying information to that agent also stops their activities.

If after some commands an agent has no active agents left who provide them with information, that agent becomes an spy themselves (in particular, if only the head remains in the group, they are considered an spy).

Leia wants to send a stop command in such a way that exactly k agents stop their intelligence activities and the final number of x spies remaining in the group is minimized.

Your task is to determine the required minimum value of x.

Input

The first line of input contains two integers n and k ($0 \le k < n \le 10^6$) — the number of agents in the group and the number of agents to whom Leia plans to send the conservation order.

The i-th of the following $n - 1$ lines contains one integer p_i — the number of the agent to whom agent i passes information ($0 \le p_i < i$). The agent with number 0 is always the head of the group.

Output

Output a single integer — the minimum number of spies that will remain in the group after the conservation order is executed, which stops intelligence activities for exactly k agents.

Examples

standard input	standard output
5 2 0 1 0 1	1
9 3 0 0 0 1 1 5 4 6	2

Problem G. Governor Scroogius

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

Scroogius, the governor of some far province in the ancient Rome received an urgent request from local authorities for funding intelligence activities to prevent the rebellion. The request was formatted as a long string of Roman numerals, without spaces.

Governor then wrote the request to Caesar, having previously placed spaces in the request so that the string represented a sequence of correctly written Roman numbers with the maximum possible sum. After receiving the requested money, the governor arranged the spaces in the original request string so that the string represented a sequence of correctly written Roman numbers with the minimum possible sum (the number of spaces in the first and second arrangements could differ), and sent this amount of money to the local authorities along with the new decryption of the string. Legend says that Scroogius profit from this venture is exactly n.

Unluckly, the legend does not tell the exact string of the Roman numerals, but tells only value of n. Check if such a situation is possible, and if so, what could the shortest string look like from which such a story could have occurred.

Your task is to answer this question for a given value of n. Moreover, the string you must output should be lexicographically smallest.

We remind you of the way Roman numbers are written (the table is taken from Wikipedia):

Place Value	Thousands	Hundreds	Tens	Units
1	M	C	X	I
2	MM	CC	XX	II
3	MMM	CCC	XXX	III
4		CD	XL	IV
5		D	L	V
6		DC	LX	VI
7		DCC	LXX	VII
8		DCCC	LXXX	VIII
9		CM	XC	IX

Note that:

- The numbers 4, 9, 40, 90, 400, and 900 are written in reverse notation, where the first symbol is subtracted from the second (for example, for 40 (`XL`), '`X`' (10) is subtracted from '`L`' (50)). This is the **only** place where reverse notation is used.
- A number containing multiple decimal digits is constructed by appending the Roman equivalent of each digit from the most significant to the least significant.
- If the decimal place is 0, no digits are written in that place in the Roman representation.
- The largest number that can be represented in the Roman numeral system is 3,999 (MMMCMXCIX).

Input

The first line of the input contains a single integer n — Governor's profit ($0 \le n \le 10^5$).

Output

If it is impossible to achieve the difference of n for any string, output -1. Otherwise, output the shortest string composed of the letters `I`, `V`, `X`, `L`, `C`, `D`, and `M`, for which the difference between Stirlitz's partition and Aisman's partition is exactly n. If there are multiple such strings, output the lexicographically smallest one.

Examples

standard input	standard output
0	C
1	-1

Problem H. Headquarter Reforms

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

New CEO of the MacroHard corporation issued a special order stating that the salaries of the employees of the headquarter would be paid based on bonuses.

To calculate the bonus received by an employee, three sequences of length n are used:

- $a = (a_1, a_2, \ldots, a_N)$ — non-negative integers defining the base bonus points.
- $b = (b_1, b_2, \ldots, b_N)$ — non-negative integers used to modify the bonus.
- $s = (s_1, s_2, \ldots, s_N)$ — a sequence of 0s and 1s representing information about the employee's presence at the workplace (1 — the employee was present, 0 — not present).

The bonus $P(\ell, r)$ received by an employee for the period from the starting day ℓ to the ending day r (where $1 \le \ell \le r \le n$) is calculated as follows:

$$P(\ell, r) = g(\ell, \ell) + g(\ell, \ell + 1) + \cdots + g(\ell, r - 1) + g(\ell, r)$$

where

$$g(\ell, i) = (A_i + B_i \times f(\ell, i)) \times S_i$$

The function $f(\ell, i)$ is defined as follows:

- If $S_i = 0$, then $f(\ell, i) = 0$.
- If $S_i = 1$, then $f(\ell, i)$ is the largest integer t satisfying the conditions:
  1. $\ell \le i - t + 1$
  2. All elements of the sequence S from the $(i - t + 1)$-th to the i-th are equal to 1 (i.e., the employee was present at work for t consecutive days up to day i, starting from a day no earlier than ℓ).

You are tasked with verifying whether the bonus point calculation mechanism works correctly. To do this, you need to write a program that processes q queries. Each query consists of three elements (c, ℓ, r) and indicates the following action:

- If $c = 1$: invert the values in the sequence S from the ℓ-th to the r-th element (i.e., replace 0 with 1 and 1 with 0).
- If $c = 2$: calculate the bonus $P(\ell, r)$ and output the result.

Can you create a program that efficiently processes the queries?

Input

The first line of the input contains the length of the sequences n ($1 \le n \le 10^5$) and the number of queries q ($1 \le q \le 10^5$).

The second line contains n elements of the sequence a ($0 \le a_i \le 10^7$).

The third line contains n elements of the sequence b ($0 \le b_i \le 10^7$).

The fourth line contains n elements of the sequence s ($s_i \in \{0, 1\}$).

The subsequent q lines contain three numbers each: c_i ($c_i \in \{1, 2\}$), ℓ_i, r_i ($1 \le \ell_i \le r_i \le N$). It is guaranteed that there is at least one query with $c_i = 2$.

Output

Process the queries in the order they are received. For each query with $c_i = 2$, output the result of the calculation $P(\ell_i, r_i)$ on a separate line.

Example

standard input	standard output
6 5 1 2 3 5 4 10 9 4 5 3 2 1 0 0 0 1 1 1 2 1 6 1 1 4 2 1 6 2 2 6 2 3 4	29 56 37 8

Problem I. Infinite Game

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

Two players are playing the following game: initially, a string of zeros, ones, and twos is written on the board. Players take turns appending one digit from the set $\{0, 1, 2\}$ to the right.

If after a player's move the sum of the written digits becomes divisible by 3 or the string ends with two identical digits in a row, that player is declared the loser (and their opponent is the winner).

Assuming both players play optimally and strive to win as quickly as possible (if such an opportunity exists), it is required to determine who will win after n moves in optimal play — the first player or the second. If at the moment of n moves the game is not over, output the remainder of the number of possible games of n moves modulo 998 244 353.

Input

The first line contains a non-empty string of digits from 0 to 2. The length of the string does not exceed 10^4 characters.

The second line contains one integer n ($1 \le n \le 10^6$).

Output

If at the moment of the n-th move the first player wins, output `-1`. If at the moment of the n-th move the second player wins, output `-2`. If at the moment of the n-th move the game is still ongoing, output a single integer — the remainder of the number of possible games from n moves modulo 998 244 353.

Examples

standard input	standard output
120 1	2
1 2	1

Problem J. Just One Attempt...

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

Thisis an interactive problem

...In the Galaxy Far Far Away, the Emperor tried to open his safe and realized that he had forgotten the current password. The password was a Fibonacci number not exceeding 2^{61}. As a hint, the sum of the digits of the number was provided. Additionally, there were two buttons on the safe: `+` and `-`. The first button would give the sum of the digits of the next Fibonacci number, while the second would give the sum of the digits of the previous Fibonacci number. However, if the previous number does not exist or if the next number exceeds 2^{61}, the contents of the safe would be destroyed. After pressing exactly one of the buttons, there would be one last attempt to recover the original password.

The Emperor realized that he was in a hopeless situation. There was one last chance — Darth Vader surely must have known this password. He had to go to Darth Vader and ask for help. Darth Vader looked at the hint, pressed one of the buttons, and then entered the password. The safe opened.

In fact, Darth Vader did not know the password – he simply recovered it. But will two hints be enough for you?

In this problem, the Fibonacci sequence is constructed as follows: $f_0 = 1$, $f_1 = 2$, $f_i = f_{i-1} + f_{i-2}$ for $i \geq 2$.

Interaction Protocol

The interaction starts with the jury program outputting a single integer — the sum of the digits in the decimal representation of the hidden Fibonacci number $f(x)$. After that, you can output one of two strings — `+` or `-`. When you output `+`, the jury program will return the sum of the digits in the decimal representation of the number $f(x+1)$. If $f(x+1) > 2^{61}$, you will receive a Wrong Answer verdict. When you output `-`, the jury program will return the sum of the digits in the decimal representation of the number $f(x-1)$. If $x = 0$, you will also receive a Wrong Answer verdict. After this, you must output the value of $f(x)$.

Do not forget to output a newline character after the commands `+` and `-`, as well as after outputting the answer, and also flush the input-output buffer by calling the `flush` function of the programming language you are using.

Example

standard input	standard output
2 3	 + 2

Problem K. Kenobi and Skywalker

Input file:	`standard input`
Output file:	`standard output`
Time limit:	3 seconds
Memory limit:	256 megabytes

The Jedi wannabe Luke found an array of n elements, whose values are positive integers from 1 to 511.

Luke receives q queries from Obi-Wan Kenobi. The i-th query is described by a pair (x_i, c_i). Luke must answer "Yes" in response if there exists a subset of the array with c elements such that its xor equals x. Otherwise, Luke must answer "No". In the query, x_i is a non-negative integer, and c_i is a natural number.

Your task is to write a program that helps Luke to answer the queries.

Input

In the first line, you are given two integers n and q ($1 \le n \le 10^6$, $1 \le q \le 10^6$), denoting the number of elements in the array and the number of queries. In the next line, you are given the elements of the array a_i – non-negative integers from 1 to 511. In the following q lines, you are given the queries. The i-th query is described by a triplet (x_i, c_i), $0 \le x_i \le 511$, $1 \le c_i \le n$, with the numbers in the query being integers.

Output

For each query, output the answer in a separate line – "Yes" or "No".

Example

standard input	standard output
5 6 3 1 2 2 3 3 1 3 2 3 5 1 4 1 5 2 5	Yes Yes No No Yes No

Problem L. Limit The Arrow

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

The archer is located at the point with coordinate 0 and shoot in the direction of increasing the Ox axis. The arrow is shot with a speed of p. At the point with coordinate i for $1 \le i \le n$, a target of thickness a_i can be installed (or there may be no target at the corresponding point). When the arrow hits a target, if the speed of the arrow is strictly greater than the thickness of the target, it decreases by a_i, otherwise, the arrow gets stuck in the target.

For each integer i from 1 to n, we define $f(i)$ as the minimum number of targets that must be installed for the arrow to travel a distance not exceeding i (i.e., getting stuck in a target at position i or at a position with a smaller number). Your task is to compute the values of $f(i)$ for all numbers from 1 to n (if for any set of installed targets the arrow flies beyond position i, then $f(i) = -1$).

Input

The first line of input contains two integers n and p ($1 \le n \le 5 \cdot 10^5$, $1 \le p \le 10^9$) — the number of points where targets can potentially be installed, and the initial speed of the arrow, respectively.

The second line contains i integers a_i ($1 \le a_i \le 10^9$). The i-th of these numbers is the thickness of the target that can be installed at point i.

Output

Output n integers $f(i)$, where $f(i)$ is the minimum number of installed targets that stop the arrow at a distance of i or less from the shooting point. If the arrow does not stop at the corresponding distance for any number of targets, output -1.

Example

standard input	standard output
5 11 4 5 2 2 11	-1 -1 3 3 1

Problem M. Minimum Number of Tasks

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

In individual programming competitions, tasks are designated by consecutive uppercase Latin letters, starting from `A`, in order of increasing difficulty (according to the jury) (the simplest task is `A`, the next one is `B`, and so on). Thus, there can be no more than 26 tasks in a round.

One participant recorded the letters corresponding to the tasks for which he submitted solutions during the competition. Since he did not always submit tasks on the first attempt, the corresponding letters could repeat; for example, the record `ABCBCE` indicates that he made an attempt for A, then an attempt for B, then an attempt for C, then made another attempt for B, then made another attempt for C, and finally made an attempt for E.

After leaving the tournament, he took the record with him but forgot to take the task conditions. Moreover, it turned out that he forgot how many tasks were in the round...

Based on the given record, restore the minimum number of tasks that could have been in the round.

Input

The first line contains a string consisting of at least one and at most 100 lowercase Latin letters.

Output

Output a single integer – the smallest possible number of tasks in the set.

Examples

standard input	standard output
GOOD	15
LUCK	21
HAVE	22
FUN	21